

Weitian Lu

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EDUCATION

University of Edinburgh **2021.09 - 2022.09**
MSc in Structural and Fire Safety Engineering Edinburgh, UK
- Core Modules: Burning and Fire dynamic, Numerical Modelling (FDS & FEM), Fire Safety Assessment

Beijing Institute of Technology, Zhuhai **2016.09 - 2020.09**
B.S. in Civil Engineering Zhuhai, Guangdong, China
- Core Modules: Structural Mechanics, Material Mechanics, Building Information Model (BIM)

WORK EXPERIENCES

Guangzhou Building Materials Institute Limited Company **2023.09-Current**
Researcher Guangzhou, China
- Complete fire experiments design and budgeting for special constructions.
- Conduct fire tests for structural elements and upholstery furniture.
- Analyze data on fire tests and wrote reports.

ZhongFang Building Construction Company **2020.10-2021.04**
Constructor Zhaoqing, China
- Performed building measurement.
- Conceived construction schemes.
- Compiled engineering plans in various aspects, including engineering mechanics calculations.

PROFESSIONAL PROJECTS

Research on CBUF Heat Release Rate Predictive Model for Mattress **2024.02-2024.04**
Project Leader
- Introduced an improved predictive method for thick mattress HRR based on the CBUF model.
- Undertook cone calorimeter and furniture calorimeter experiments and obtained input data of the model.
- Applied computer vision technology for flame area calculation, and compared predictive and experimental results.
- **Output**: *An improved HRR predictive method for mattress based on CBUF model* (Chinese version, accepted in The Third National Symposium on Thermal Safety Science and Technology)

Improved Fire Testing Method for Floor Coverings based on Computer Vision Technology **2023.11-2024.01**
Project Leader
- Solved view limited problems in instrument through perspective transformation.
- Developed an automated detection method for flame front location through color conversion and statistics.
- **Output:** An enhanced automated fire testing software for floor coverings.

Research on A Reduced-scale Test Method for One-way Slab of Metro **2023.11**
Associated Researcher
- Assisted with conceptualization, proposing a reduced-scale fire test method for evaluating the fire resistant time of one-way slab.
- Conducted full and reduced-scale experiments, and processed experimental data.
- Reviewed literature and wrote the research paper.
- **Output:** *Research on the Scale Fire Test and Fire Resistance of the One-Way Slab of a Metro* (Qiu et al., 2024).

Building Collapse by Fire Experiment – Experimental Procedure Design **2023.10-2023.11**
Associated Researcher
- Calculated the required heat release rate for floor slab collapse.
- Devised the ventilation condition to allow ventilation-controlled burning.
- Estimated the time when the floor slab collapsed, and verified 4-minute lag from experiment results.
- **Output:** Experimental procedure scheme

Heat Flux Analysis in Experiment and FDS Model for Travelling Fires **2022.03-2022.09**
- Pre-calculated the data measured from thin skin calorimeter.
- Analyzed convective heat transfer leading to heat flux and fire spread.
- Extracted heat flux distribution from the calibrated FDS model.
- **Output:** Dissertation, heat flux contour maps for FDS model, adopted in *"Scaling-up" fire spread on wood cribs to predict a large-scale travelling fire test using CFD* (Dai et al., 2024)

PUBLICATIONS

- Qiu, P., Duan, J., Yang, Z., Liu, J., **Lu, W.,** 2024. Research on the Scale Fire Test and Fire Resistance of the One-Way Slab of a Metro. *Buildings* 14, 1695.
- **Lu, W.**, Liu, J., Duan, J., Gao, S., Wu X., 2024. An improved HRR predictive method for mattress based on CBUF model. *The Third National Symposium on Thermal Safety Science and Technology*

AWARDS

- Outstanding undergraduate graduation thesis **(5/180)**
- College Scholarships (for six semesters)

SKILL SET

- **Fire Analysis:** Fire & Burning Dynamic, Heat Transfer, Fire Dynamic Simulator (FDS)
- **Image Processing Techniques:** OpenCV, Open3D
- **Data Processing and Visualization:** Python, Matlab, Origin
- **Deep Learning Techniques:** PyTorch